

On Butler's Positivity Conjecture

(joint with P. Guo and M. Kang)

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Macdonald Polynomials

Schur [1901]

$$s_\lambda(X) \in \Lambda$$

Schur function



Littlewood [1961]

$$P_\lambda(X; t) \in \Lambda_t$$

Hall–Littlewood
symmetric function



Macdonald [1988]

$$P_\lambda(X; q, t) \in \Lambda_{q,t}$$

Macdonald
symmetric function

There are different forms of
Macdonald functions

$$P_\lambda = m_\lambda + (\text{lower term})$$

$$J_\lambda = P_\lambda \cdot \prod_{\square \in \lambda} (1 - q^{a(\square)} t^{\ell(\square)+1})$$

$$H_\lambda = J_\lambda \left[\frac{X}{1-t} \right] = J_\lambda \begin{pmatrix} x_1 & tx_1 & t^2x_1 & \dots \\ x_2 & tx_2 & t^2x_2 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}$$

$$\tilde{H}_\lambda = t^{n(\lambda)} H_\lambda|_{t \mapsto t^{-1}}$$

Theorem (Haiman, 2001)

$$\tilde{H}_\lambda \in \sum \mathbb{N}[q, t] \cdot s_\theta.$$

This was conjectured by Macdonald.

Example

$$\tilde{H}_{(1)} = s_{(1)}$$

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \square & 0 & 0 \end{array}$$

$$\tilde{H}_{(2)} = s_{(2)} + qs_{(1,1)}$$

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \square\square & \begin{array}{c} \square \\ \square \end{array} & 0 \end{array}$$

$$\tilde{H}_{(2)} = s_{(2)} + ts_{(1,1)}$$

$$\begin{array}{ccc} 0 & 0 & 0 \\ \begin{array}{c} \square \\ \square \end{array} & 0 & 0 \\ \square\square & 0 & 0 \end{array}$$

$$\tilde{H}_{(3)}$$

$$\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \square\square\square & \begin{array}{c} \square \\ \square \end{array} & \begin{array}{c} \square \\ \square \end{array} & \begin{array}{c} \square \\ \square \\ \square \end{array} & 0 \end{array}$$

$$\tilde{H}_{(2,1)}$$

$$\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \begin{array}{c} \square \\ \square \end{array} & \begin{array}{c} \square \\ \square \\ \square \end{array} & 0 & 0 & 0 \\ \square\square\square & \begin{array}{c} \square \\ \square \end{array} & 0 & 0 & 0 \end{array}$$

$$\tilde{H}_{(1,1,1)}$$

$$\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ \begin{array}{c} \square \\ \square \\ \square \end{array} & 0 & 0 & 0 & 0 \\ \begin{array}{c} \square \\ \square \end{array} & 0 & 0 & 0 & 0 \\ \begin{array}{c} \square \\ \square \end{array} & 0 & 0 & 0 & 0 \\ \square\square\square & 0 & 0 & 0 & 0 \end{array}$$

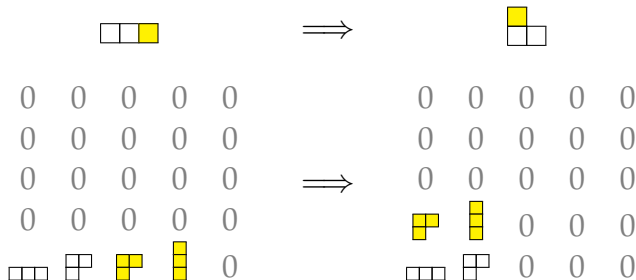
Butler's conjecture

Observing from the computation, Butler conjectured that

if μ is obtained by moving a box of λ , then

$\tilde{H}_\mu$ is obtained by moving a summand of $\tilde{H}_\lambda$.

For example,



Butler's conjecture

Note that if $\mu \vdash n$,

$$\tilde{H}_\mu = 1 \cdot h_n + \cdots + T_\mu \cdot e_n,$$

where $T_\mu = \prod_{(i,j) \in \mu} q^i t^j = q^{n(\mu')} t^{n(\mu)}$. The precise conjecture can be stated as follows. we can write

$$\begin{aligned}\tilde{H}_\lambda(X; q, t) &= 1 \cdot I_{\lambda, \mu}(X; q, t) + T_\lambda \cdot J_{\lambda, \mu}(X; q, t) \\ \tilde{H}_\mu(X; q, t) &= 1 \cdot I_{\lambda, \mu}(X; q, t) + T_\mu \cdot J_{\lambda, \mu}(X; q, t),\end{aligned}$$

such that

$$I_{\lambda, \mu}(X; q, t), J_{\lambda, \mu}(X; q, t) \in \sum \mathbb{N}[q, t] \cdot s_\theta(X).$$

Butler's conjecture

Since $T_\lambda \neq T_\mu$ if $\lambda \neq \mu$, we have

$$l_{\lambda,\mu}(X; q, t) = \frac{T_\lambda \tilde{H}_\mu(X; q, t) - T_\mu \tilde{H}_\lambda(X; q, t)}{T_\lambda - T_\mu}. \quad (*)$$

Using the reciprocity identity:

$$\tilde{H}_\mu(X; q, t) = t^{n(\mu)} q^{n(\mu')} \omega \tilde{H}_\mu(X; q^{-1}, t^{-1}),$$

it is easy to see $J_{\lambda,\mu}(X; q, t) = \omega l_{\lambda,\mu}(X; q^{-1}, t^{-1})$. So the conjecture is equivalent to

$$(*) \in \mathbb{N}[q, t] \cdot s_\theta(X).$$

Geometry of Hilbert schemes

Recall

$$\begin{aligned} H_n &= \text{Hilbert scheme of } n\text{-points in } \mathbb{C}^2 \\ &= \{I \trianglelefteq \mathbb{C}[x, y] : \dim_{\mathbb{C}} \mathbb{C}[x, y]/I = n\}. \end{aligned}$$

By Fogarty, H_n is a nonsingular variety of dimension $2n$.

Example

When $n = 1$, any $I \in H_n$ is a maximal ideal, i.e. it takes the form $\langle x - x_0, y - y_0 \rangle$ for some $(x_0, y_0) \in \mathbb{C}^2$, so

$$H_n \cong \mathbb{C}^2.$$

Example

When $n = 2$, it is known that

$$H_n \cong \mathbb{C}^2 \times T^*\mathbb{P}^1$$

where $\mathbb{P}^1$ corresponds to I with $\mathbb{C}[x, y]/I \cong \mathbb{C}[\epsilon]/\epsilon^2$.

Geometry of Hilbert schemes

Consider

$$\begin{aligned}\mathbb{C}^{2n}/S_n &= \operatorname{Spec} \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]^{S_n} \\ &= \{\text{multi-sets of } n\text{-points in } \mathbb{C}^2\}.\end{aligned}$$

We have Hilbert–Chow map $\sigma : H_n \longrightarrow \mathbb{C}^{2n}/S_n$ by sending I to the support (with multiplicity) of $\mathbb{C}[x, y]/I$.

Example

When $n = 1$, a multi-set of one point is just one point, so

$$\mathbb{C}^{2n}/S_n = \mathbb{C}^2.$$

Then σ is identical.

Example

When $n = 2$, by linear transformation, we have

$$\mathbb{C}^{2n}/S_n \cong \mathbb{C}^2 \times (\mathbb{C}^2/\{\pm 1\}).$$

Then σ is a blow-up map.

Geometry of Hilbert schemes

The **isospectral Hilbert scheme** X_n is defined to be the reduced fiber product in the following diagram

$$\begin{array}{ccc} X_n & \longrightarrow & \mathbb{C}^{2n} \\ \rho \downarrow & & \downarrow \\ H_n & \xrightarrow{\sigma} & \mathbb{C}^{2n} / S_n. \end{array}$$

Theorem (Haiman)

The map ρ is finite and flat.

The **Procesi bundle** is the sheaf

$$\mathcal{P} = \rho_* \mathcal{O}_{X_n}.$$

This is a $(T \times S_n)$ -equivariant vector bundle, where $T = \mathbb{C}^\times \times \mathbb{C}^\times$ acts naturally on H_n , and S_n acts trivially.

Example ($n = 2$)

When $n = 2$, we can identify H_n with the variety of

- ▶ two distinct **unordered** points in $\mathbb{C}^2$;
- ▶ a point with a tangent direction in $\mathbb{C}^2$.

We can identify $H_n = \mathbb{C}^2 \times \mathbb{A}(\mathcal{O}_{\mathbb{P}^1}(-2))$.

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The map $\rho : X_n \rightarrow H_n$ is by taking square

$$sq : \mathbb{C}^2 \times \mathbb{A}(\mathcal{O}_{\mathbb{P}^1}(-1)) \rightarrow \mathbb{C}^2 \times \mathbb{A}(\mathcal{O}_{\mathbb{P}^1}(-2)).$$

Example ($n = 2$)

Let $L \subset \mathbb{C}^2$ be an element of $\mathbb{P}^1$. The map on each fiber is

$$sq : L \longrightarrow L^{\otimes 2}.$$

Choosing a nonzero $z \in L^*$, we have

$$\mathbb{C}[L] = \mathbb{C}[z], \quad \mathbb{C}[L^{\otimes 2}] = \mathbb{C}[z^2].$$

The $-1 \in S_2$ acts by $z \mapsto -z$. Note that the zero of $L^{\otimes}$ is defined by $z^2 = 0$. In particular, we have

$$sq_* \mathcal{O}_L|_{\{0\}} = \mathbb{C}[z] / \langle z^2 \rangle \cong \mathbb{C} \oplus \mathbb{C}z = \mathbb{C} \oplus L^{\vee}.$$

The above computation glues to the following identification

$$\mathcal{P}|_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1} \otimes \text{tri} \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes \text{sgn}.$$

The connection

The isomorphism we computed in $n = 2$

$$\mathcal{P}|_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1} \otimes \text{tri} \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes \text{sgn}$$

is compatible with

$$\begin{aligned}\tilde{H}_{(2)} &= 1 \cdot s_{(2)} + q \cdot s_{(1,1)}, \\ \tilde{H}_{(1,1)} &= 1 \cdot s_{(2)} + t \cdot s_{(1,1)}.\end{aligned}$$

This is obtained by localization and taking Frobenius character.

The connection

Let $\mu \vdash$ be a partition. We can define a monomial ideal

$$I_\mu = \langle x^a y^b : (a, b) \notin \mu \rangle \in H_n$$

$$\mu = \begin{array}{|c|c|c|c|} \hline y^3 & & & \\ \hline & xy^3 & & \\ \hline & & x^2y & \\ \hline & & & x^4 \\ \hline \end{array}$$

$$I_\mu = \langle x^4, x^2y, xy^3, y^3 \rangle.$$

Theorem (Haiman)

$$\text{Frob}_{q,t}(\mathcal{P}|_{I_\mu}) = \tilde{H}_\mu.$$

Here $\text{Frob}_{q,t}$ is the bi-graded Frobenius character

$$\{(T \times S_n)\text{-representations}\} \xrightarrow{\text{Frob}_{q,t}} \sum_{\theta} \mathbb{N}[q^{\pm 1}, t^{\pm 1}] s_{\theta}$$

The connection

Sketch of the proof of Butler's conjecture.

Let λ, μ differ by a box. Then we can find a T -equivariant curve $f : \mathbb{P}^1 \xrightarrow{\subseteq} H_n$ joining l_λ and l_μ .

Now by Birkoff–Grothendieck theorem and some properties of $\mathcal{P}$, we can decompose

$$f^*\mathcal{P} \cong \mathcal{O}_{\mathbb{P}^1} \otimes M \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes N.$$

The weight of $\mathcal{O}_{\mathbb{P}^1}(1)|_{l_\mu}$ is T_μ . We thus have

$$\tilde{H}_\lambda = 1 \cdot \text{Frob}_{q,t}(M) + T_\lambda \cdot \text{Frob}_{q,t}(N),$$

$$\tilde{H}_\mu = 1 \cdot \text{Frob}_{q,t}(M) + T_\mu \cdot \text{Frob}_{q,t}(N).$$

We are done.

Thanks